HSING-YU (WINNIE) WEN

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA — Expected Dec 2026
Master of Information Systems Management – Business Intelligence & Data Analytics

Soochow University, Taipei, Taiwan — Sep 2019 – Jun 2023
BBA in International Business | GPA 4.00/4.00 | Rank 1/68 in class, 1/136 in department

TECHNICAL SKILLS

Languages & ML: Python, SQL, Java, PyTorch, TensorFlow, scikit-learn, Spark, NLP, Git
GenAI & Retrieval: LLMs, RAG, Hybrid Semantic + Keyword Retrieval, Vector Databases (Qdrant), MCP, LangChain, LangGraph
MLOps & Cloud: Model Serving, Monitoring, Drift Detection, Kafka, Docker, Kubernetes, Helm, GitHub Actions, AWS, GCP, Azure

WORK EXPERIENCE

Sustainable Future Tech | ***Forward-Deployed Engineer (CMU Capstone)***, Pittsburgh, PA — Sep 2026 – Present
- Building schema-bound LLM Detection and Action Proposal agents that turn security telemetry into validated risk contracts and containment proposals for an AI-driven SOC workflow, with JSON Schema validation.
- Integrating the agent pipeline with the sponsor's AI Governance Control Plane (AGCP v2) REST API so containment actions execute only after runtime authorization at a Policy Enforcement Point (PEP).
- Running replayable baseline-versus-governed experiments with fail-closed and bypass testing and auditable evidence.

TSMC | ***Data AI Engineer Intern***, Hsinchu, Taiwan — May 2026 – Aug 2026
- Engineered an enterprise RAG knowledge platform indexing 10,000+ Power BI reports on MariaDB and Qdrant, combining semantic and keyword retrieval for natural-language analytics and achieving 90% HitRate@5.
- Automated metadata standardization with an LLM-powered pipeline that transforms Power BI REST API outputs into standardized, AI-ready knowledge files, saving an estimated 1,200+ hours of manual documentation.
- Implemented an MCP service mapping natural-language questions to report and dataset IDs for source-aware DAX generation.

Bank SinoPac | ***Data Analyst***, Taipei, Taiwan — Jun 2024 – Jun 2025
- Optimized customer targeting with uplift modeling and A/B testing, increasing marketing ROI by 25% and guiding leadership.
- Delivered 20+ analytics products with Python, T-SQL, and Tableau, improving decision workflows by 47% across digital banking.
- Led analysis for the 2025 flagship O2O initiative, identifying high-potential customers and generating NT$5–10M monthly.

ASUS | ***Data Analyst Intern***, Taipei, Taiwan — May 2023 – Nov 2023
- Modeled industry-specific content preferences with logistic regression, increasing conversion by 31% versus the prior campaign.
- Partnered with Deloitte on a Salesforce CRM rollout and analytical dashboards, accelerating campaign optimization by 67%.

GoFreight | ***Customer Success Intern***, Taipei, Taiwan — Dec 2021 – Jul 2022
- Forecasted demand with Python and Excel time-series models, improving staffing decisions and reducing personnel costs by 21%.
- Diagnosed system and crawler failures in container-tracking data, raising accuracy to 98% and cutting manual processing by 20%.

PROJECTS

Banking Text-to-SQL RAG | *LLM, FAISS, Sentence-Transformers, SQLite* — May 2026
- Built a Text-to-SQL RAG pipeline that retrieves schema chunks from a FAISS index, expands context through foreign-key relationships, and prompts an LLM to generate SQLite queries for non-technical business users.
- Added out-of-scope detection, read-only SQL validation, and a deterministic evaluator that checks results against gold SQL.

Movie Recommendation System | *Kafka, Docker, MLOps* — Jan 2026 – Apr 2026
- Deployed a Kafka-based recommendation pipeline with a containerized hybrid inference API serving at sub-600 ms latency.
- Established data-quality checks, tests, monitoring, drift detection, and offline/online evaluation for recommendation serving.

Heterogeneous Cloud Storage Backend | *Java, MySQL, Neo4j, MongoDB, Azure, Terraform, Ansible* — Oct 2026
- Built a Java backend for a social-network timeline that fans out across MySQL, Neo4j, and MongoDB on Azure, serving 2.6M users and 37.6M comments (~10 GB) with compound indexes; provisioned via Terraform and Ansible.
- Cut cached top-user timeline latency from 8.28 s to 4 ms and tuned retries to eliminate failures under injected faults.

Multi-Cloud Microservice Deployment | *AWS, GKE, AKS, Terraform, Helm, GitHub Actions* — Sep 2026
- Deployed Profile, Login, and Chat microservices with MySQL and Redis to GKE and AKS using Docker, Kubernetes, and Helm.
- Automated image builds and Helm releases with GitHub Actions, and compared manual, Python API, and Terraform deployments of AWS EC2 Auto Scaling Groups behind an ALB with health checks and CloudWatch monitoring.